

PM Project 4: Final Strategy Notes

Exact rules, motivation, evidence and failure modes

Decision. Use `linear_fast_shrunk_core` for expected CAGR. Use `linear_fast_shrunk_core_survival` for heavier-tail survival and `linear_fast_shrunk_core_tstat` for permanent-reversal insurance.

1. Problem

Choose daily weights for 20 securities. Day- t targets may use only information known by day $t-1$. The score is after-cost CAGR. A security can move by at most 0.05 weight units per day; trading costs 5 bp each way; a return below -100% busts the book.

2. Signal book

Cross-sectionally standardise each signal, lag one day, then EWMA-smooth signals 1/2/3 with half-lives 5/8/3 days. Let the smoothed panels be u_1 , u_2 and u_3 .

```
s(t) = XSZ[0.30 u1(t) + 0.40 u2(t) + 0.30 u3(t)]
a_i(t) = s_i(t) / sum_k |s_k(t)|
```

The unit-gross book $a(t)$ is long high scores and short low scores. Signal 2 is weak alone, but deleting it damaged the joint book. A 30% weight gave similar full-sample growth to 40%, yet had a worse worst fold, drawdown and trading cost. This is portfolio evidence, not standalone significance.

3. Slow shrunk core

Estimate lagged EWMA mean and variance for each security with a 1.000-day half-life. Define $\alpha_i(t) = \text{mean}_i(t) / \text{variance}_i(t)$.

```
raw_i(t) = max{1 + 0.30 XSZ[q_i(t)], 0.25}
c_i(t) = raw_i(t) / sum_k raw_k(t)
```

Use equal weights for the first 500 days. The core stays long-only and close to equal weight. The 0.30 strength is rounded within a broad stable region: the floor prevents noisy estimates from deleting a security.

```
b(t) = unitgross[0.525 c(t) + 0.475 a(t)]
```

PC1 was rejected because it identifies maximum common variance, not expected return. Tested PC1 cores did not improve the relevant growth/risk trade-off.

4. Risk sizing and timing

For paper return $p(t) = b(t)r(t)$, estimate downside risk from the 60-day EWMA of $2 \min(p, 0)^2$. The factor 2 equals ordinary variance for symmetric returns.

```
L(t) = clip[0.0375 / downside_risk(t) + 0.5 timing_S2(t), 2, 6]
```

The survival-first variant changes only 0.0375 to 0.0300. It lowers ordinary expected growth but survived every path in the deliberately extreme heavy-tail boundary test; this is evidence, not a guarantee.

`timing_S2` is the cross-sectional mean of signal 2, standardised only against its past 500-day EWMA moments, clipped to $[-3, 3]$, and smoothed over 5 days. It is separate from signal 2's cross-sectional role. Removing it worsened growth, drawdown, worst-day loss and cost in the frozen comparison, but this is still in-sample evidence.

Ideal weights are $\text{clip}[L(t)b_i(t), -1.5, 1.5]$.

5. Executable targets

Drift yesterday's actual weights by realised returns and the post-cost portfolio denominator. Trade only the desired change outside a one-asset-volatility deadband. Clip planned changes to ± 0.049 and submitted gross to 9.5. The 0.049 buffer avoids routine contact with the official 0.05 cap.

6. T-stat insurance variant

Let $T(t)$ be a correctly scaled, lagged 120-day EWMA t-statistic of the realised cross-sectional IC correlations $i(t)$, $r_i(t)$.

```
lambda(t) = 0.475 tanh{2[T(t) + 1]}
b(t) = unitgross[(1 - |lambda(t)|)c(t) + lambda(t)a(t)]
```

The +1 is a prior for the original signal direction. Persistent negative evidence can short the signal book. The literal extrapolating $2 \tanh(2T)$ rule produced negative training CAGR; an additive interpretation still lost substantial growth. Multiplying a noisy t-stat creates responsiveness, not information.

7. Evidence and limits

Check	Finding	Meaning
Training	Higher CAGR/Sharpe; lower drawdown and worst-day loss.	Supports selection, but is design data.
Folds	Better worst fold; 9/10 folds positive.	No single short regime explains all growth.
Leave-one-out	Growth and selection score improved for all 20 omissions.	Not one lucky security.
Block bootstrap	About 73-79% positive draws; every 95% interval crosses zero.	Plausible, not statistically established.
Long stress	Strong under drift, alpha, volatility, tails and asset perturbations.	Model-risk evidence, not new data.
Severe tail	The 3.0% target had no busts in 20 extreme paths; higher targets failed.	Lower risk helps; ruin remains possible.

The slow quality relation is stronger in the later half. This is the main overfit concern. Use the fixed rule for expected growth, the 3.0% rule for a strong heavy-tail prior, and the t-stat rule for a strong permanent-reversal prior. No tested strategy guarantees survival under arbitrary heavy tails.

The accompanying Python files are authoritative. Editable LaTeX source is included beside this PDF.